

11 (long)

A packing crate is to be designed to transport frozen produce within a refrigerated truck. The storage volume is a cube of side length $L = 0.3$ m, which is a fixed dimension. This is enclosed on all sides by walls of uniform thickness d . The crate wall material and thickness d are free variables. A cross-section through the crate is shown in Fig. 7.

The thermal insulation is characterised by the time scale τ for heat transfer through the walls

$$\tau = \frac{d^2 \rho c}{\lambda}$$

where a larger τ indicates better insulation. When crates are stacked, the walls will buckle when the force F on a crate is

$$F = \frac{4\pi^2 E d^3}{3L}.$$

Candidate materials and their property values are given in Table 1.

(a) A material is to be selected for the crate that maximises the time constant τ , subject to the constraint $F \geq 700$ N.

(i) Derive a material performance index that should be maximised. [4]

(ii) Identify the best material in Table 1, and give the value of d . [4]

(b) To refine the selection, a material is to be chosen that minimises the wall thickness d subject to the constraints $F \geq 700$ N and $\tau \geq 1800$ s.

(i) Identify the best material in Table 1, and give the value of d . [7]

(ii) Comment on the effect of the revised selection criteria, and suggest two additional constraints that should be considered. [3]

	cork	polystyrene foam	polyethylene
Young's modulus, E (GPa)	0.032	0.28	0.90
density, ρ (kg m ⁻³)	180	270	960
thermal conductivity, λ (W m ⁻¹ K ⁻¹)	0.042	0.052	0.44
specific heat capacity, c (J kg ⁻¹ K ⁻¹)	1700	1500	1900

Table 1

(c) The crate design influences two contributions to the transport energy consumption. First, the power (in Watts) required to refrigerate the truck per crate is

$$P_r = \begin{cases} 500V \left(1 - \frac{\tau}{1800}\right) & \text{if } 0 \leq \tau \leq 1800 \text{ s} \\ 0 & \text{if } \tau > 1800 \text{ s} \end{cases}$$

where $V = (L + 2d)^3$ is the volume occupied by a crate in m^3 and τ is in seconds.

Second, the energy required to move the truck per unit mass transported, per unit distance travelled, is $0.46 \text{ kJ kg}^{-1} \text{ km}^{-1}$.

- (i) Considering only cork and polyethylene, identify the material that minimises the transport energy consumption per crate, subject to the constraint $F = 700 \text{ N}$. The lifespan of a crate is 1000 km of travel at an average speed of 40 km hr^{-1} . [9]
- (ii) Comment on how the material selection may affect the environmental impact during other stages of the crate's life-cycle. [3]

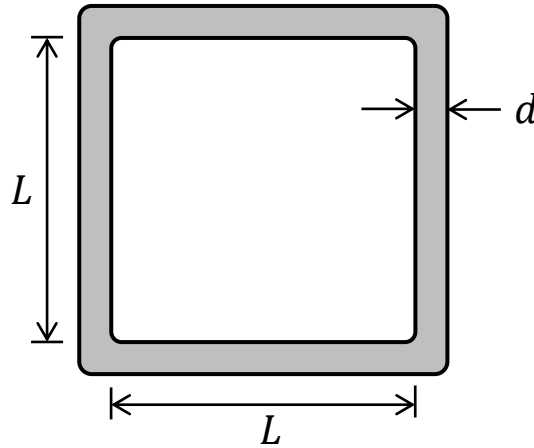


Fig. 7